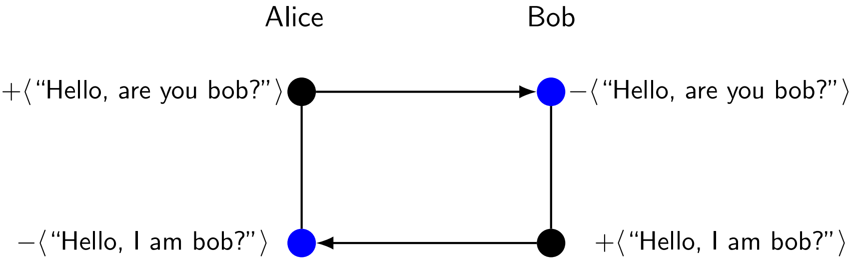
Session 3: CPSA Etymology and Attacks in CPSA

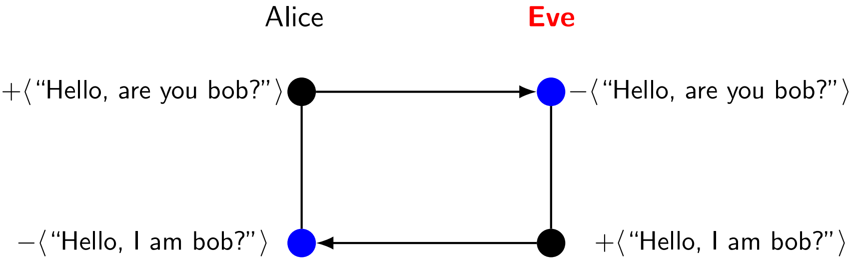
Slides prepared by UMBC Protocol Analysis Lab

Last edited June 19, 2026

A completed protocol?



Strands can lie



CPSA adversarial strand actions

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M. Text Message: $\langle +t \rangle$ where $t \in T$

T. Tee: $\langle -g, +g, +g \rangle$ where $g \in G$

C. Concatenation: $\langle -g, -h, +gh \rangle$ where $g, h, gh \in G$

S. Separation into components: $\langle -gh, +g, +h \rangle$

K. Key: $\langle +K \rangle$ where $K \in \mathbf{K_p}$

E. Encryption $\langle -K, -h, +\{h\}_K \rangle$

D. Decryption $\langle -K^{-1}, -\{h\}_K, +h \rangle$

1. $\langle +m \rangle$ emit a message from the network.
2. $\langle -m \rangle$ extract a message from the network.
3. $\langle m_1 m_2 \rangle$ concatenate two messages.
4. $\langle \{m\}_K \rangle$ encrypt a message m under key K.

An adversarial model of the network

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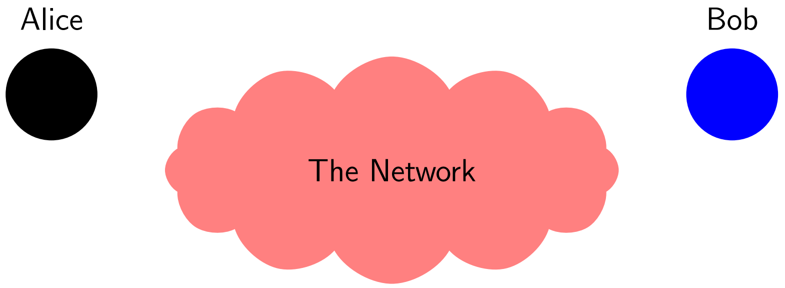
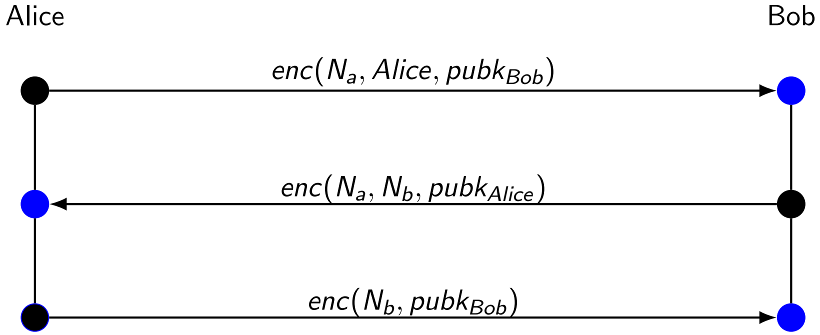


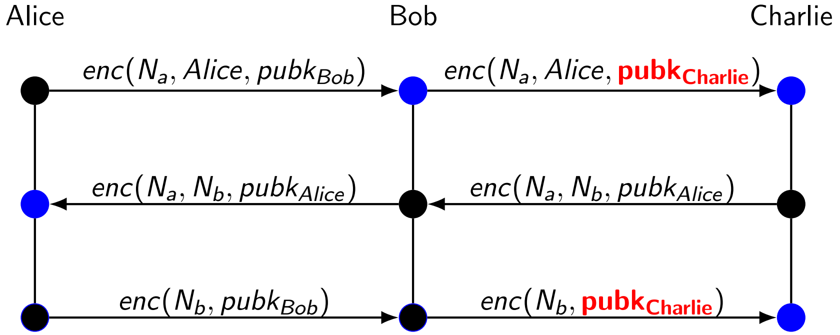
Goal: Mutually *authenticate* (verify identities) over an insecure network.

Needham Schroeder



Is this protocol secure against the Dolev-Yao Adversary?

Needham Schroeder



Remember: The Dolev-Yao Adversary can be a legitimate party on the network

What do we give CPSA?

`protocol.scm`

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- herald

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- defmacro

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What do we give CPSA?

`protocol.scm`

- herald
- defmacro
- defprotocol
  - defrole
- defskeleton
- defstrand

What does CPSA give us?

`protocol.scm` $\rightarrow$

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- `protocol.txt`

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`protocol.scm` $\rightarrow$

- `protocol.txt`
- `protocol_shapes.txt`

What does CPSA give us?

`protocol.scm` →

- `protocol.txt`
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- `protocol.xhtml`

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`protocol.scm` $\rightarrow$

- `protocol.txt`
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How can we spot attacks?

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- *SECURITY GOALS*

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- *SECURITY GOALS*
- Dotted lines
- Listener Strands

Homework

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Find attacks in CPSA

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Looking at shapes to find attacks takes some practice, so let us do some practice!

Two of the protocols are already done, you just have to run `make` so CPSA can make shapes, and then analyze them.

Two of the protocols need to be filled in, you will have to add part of a role, fill in a skeleton, or something else (documented in the comments), and then run CPSA to analyze the shapes.

Survey

Fill in details about the survey here!